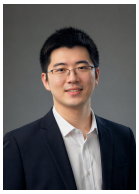
Yinda Chen 陈胤达

Ph.D. Candidate, USTC & Shanghai AI Lab
Information and Communication Engineering Expected June 2027
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Research Profile

Research focuses on embodied world models and vision-language-action (VLA) models, scientific agents and foundation-model post-training, and self-supervised multimodal representation learning. Long-term work in 3D medical imaging and connectomics addresses data-efficient learning, domain transfer, and relationships between circuit structure and function.

Education

University of Science and Technology of China & Shanghai AI Lab 2024.07–2027.06
Ph.D. in Information and Communication Engineering, joint training; expected June 2027.
Advisors: Feng Wu and Zhiwei Xiong. Connectomics research mentor: Ruobing Zhang (Fudan University).

University of Science and Technology of China 2022.09–2024.07
M.S. in Computer Technology.
Previously worked with Prof. Xiaoou Tang at Shanghai AI Lab in 2023.

Xiamen University 2018.09–2022.07
Dual bachelor's degrees in Environmental Ecological Engineering and Economics.
Undergraduate research advisor: Yuanye Zhang.

Research and Professional Experience

Beijing Innovation Center of Humanoid Robotics (X-Humanoid) 2025.12–Present
Core Member, Embodied Intelligence World Model Algorithm Team Beijing

- Core contributor to unified VLM and action modeling in Pelican-Unified 1.0, connecting vision-language understanding, future-state prediction, and robot actions.
- Contributed to task-relevant bottleneck representations and action learning in Pelican-VLA 0.5, including real-robot closed-loop validation; experience with 100-GPU H200 training clusters.

Ubiquant 2026.05–2026.10
Research Intern, Quantitative Research and Large Language Models Beijing

- Worked on quantitative research and foundation-model training, with **thousand-GPU-scale** cluster experience; explored LLM applications in factor mining and data-driven decision making.
- Served as a core contributor to **SAIL**, a scientific-agent model for literature research, scientific coding, and multi-step research workflows (2026.10).
- Contributed to **IQuest-Q1** and co-authored its technical report on agentic CLI interaction, coding, and multi-step tool use (2026.10).

Tencent Interactive Entertainment Group (IEG) 2025.08–2025.12
Research Intern, Qingyun Program Shanghai

- Worked on MAIN-VLA (Game for Peace), multimodal game world models, and Honor of Kings Lingbao TTS/live commentary using F5-TTS and IndexTTS2.
- Collaborated on router-free MoE-LoRA research in FURINA, studying merging and deployment of low-rank expert updates (IEEE IoTJ, accepted 2026).

Chinese PLA General Hospital (301 Hospital) 2023.09–2024.02
Research Intern, Data Compression Group Beijing

- Collaborated with Prof. Qionghai Dai's team on efficient medical-image compression, investigating content- and modality-specific methods for CT and MRI.

USTC | Representation Learning and Connectomics 2022.09–Present
Graduate Researcher; NSFC Doctoral Student Project PI (2025–2027) Hefei

- Research covers autoregressive pretraining (TokenUnify), multi-agent masking (dbMiM), and the theory of domain-adaptive segmentation (MaskTwins) for electron microscopy and 3D medical imaging.
- Investigated neurotransmitter lateralization and structure-function relationships in the adult Drosophila mushroom-body connectome; related abstract accepted as a poster at SfN 2026.

Imperial College London, Data Science Institute 2022.11–2023.08
Research Intern London (Remote)

- Worked with Rossella Arcucci and Che Liu on 3D medical multimodal learning: generated-text-guided vision-language pretraining (GTGM) and CT text-image retrieval (BIMCV-R / MedFinder).
- Contributed to BIMCV-R, a retrieval dataset containing 8,069 CT volumes, and the associated retrieval framework.

Xiamen University WISER Club 2021.08–2022.07
Academic Community Member, Data Mining Group Xiamen

- Designed and taught course material on clustering and Transformers, mentored undergraduate machine-learning projects, and organized campus competitions.

Wang Yanan Institute for Studies in Economics, Xiamen University 2020.08–2021.12
Research Assistant, Econometrics Xiamen

- Assisted Prof. Jiong Zhu with territorial economic statistics and homestead analysis; extracted visual features from satellite imagery to study land-use changes.

Technical Reports

[1] **IQuest-Q1: Advancing Foundation Capabilities for Agentic CLI Systems**
IQuest Research; contributor and co-author. *Technical report, 2026.10.*

[2] **SAIL: Scientific Agentic Intelligence via a Science-Aware Loop**
SAIL Model Team / IQuest Research; core contributor. *Technical report, 2026.10.*

[3] **Pelican-VLA 0.5: Attending Before Acting Benefits Generalization**
X-Humanoid, WFM System Group; contributor. *Technical report, 2026.07.*

[4] **Pelican-Unified 1.0: A Unified Embodied Intelligence Model (UEI) for Understanding, Reasoning, Imagination and Action**
X-Humanoid, WFM System Group; core contributor to unified VLM and action modeling. *Technical report, 2026.05.*

Selected Peer-Reviewed Publications

* denotes equal contribution. Authors are listed in paper order. Full list: Google Scholar.

[1] **Learned Image Coding with Generative Reference of Conditional Latents**
Siqi Wu*, **Yinda Chen***, Weiming Chen, Dong Liu, K. C. Ho, Zhihai He. *IEEE TPAMI, 2026 (accepted).*

[2] **FURINA: Free from Unmergeable Router via LINear Aggregation of mixed experts**
Jiayi Han et al. (including **Yinda Chen**). *IEEE Internet of Things Journal, 2026 (accepted).*

[3] **Joint Semantic and Coded Generation for Conditional Latent Coding**
Siqi Wu, Weiming Chen, **Yinda Chen**, Dong Liu, K. C. Ho, Zhihai He. *IEEE TCSVT, 2026.*

[4] **TokenUnify: Scaling Up Autoregressive Pretraining for Neuron Segmentation**
Yinda Chen*, Haoyuan Shi*, Xiaoyu Liu, Te Shi, Ruobing Zhang, Dong Liu, Zhiwei Xiong, Feng Wu. *ICCV, 2025.*

[5] **MaskTwins: Dual-form Complementary Masking for Domain-Adaptive Image Segmentation**
Jiawen Wang, **Yinda Chen**, Xiaoyu Liu, Che Liu, Dong Liu, Jianqing Gao, Zhiwei Xiong. *ICML, 2025; theory contribution.*

[6] **Conditional Latent Coding with Learnable Synthesized Reference for Deep Image Compression**
Siqi Wu*, **Yinda Chen***, Dong Liu, Zhihai He. *AAAI, 2025.*

[7] **GTGM: Generative Text-Guided 3D Vision-Language Pretraining for Medical Image Segmentation**
Yinda Chen*, Che Liu*, Wei Huang, Xiaoyu Liu, Haoyuan Shi, Sibo Cheng, Rossella Arcucci, Zhiwei Xiong. *ICCV Workshop (VLM3D), 2025.*

[8] **EMPOWER: Evolutionary Medical Prompt Optimization With Reinforcement Learning**
Yinda Chen*, Yangfan He*, Jing Yang, Dapeng Zhang, Zhenlong Yuan, Muhammad Attique Khan, Jamel Baili, Por Lip Yee. *IEEE JBHI, 2025 (Early Access).*

[9] **BIMCV-R: A Landmark Dataset for 3D CT Text-Image Retrieval**
Yinda Chen, Che Liu, Xiaoyu Liu, Rossella Arcucci, Zhiwei Xiong. *MICCAI, 2024.*

[10] **MaskFactory: Towards High-quality Synthetic Data Generation for Dichotomous Image Segmentation**
Haotian Qian*, **Yinda Chen***, Shengtao Lou, Fahad Shahbaz Khan, Xiaogang Jin, Deng-Ping Fan. *NeurIPS, 2024.*

[11] **Unsupervised Domain Adaptation for EM Image Denoising With Invertible Networks**
Shiyu Deng, **Yinda Chen**, Wei Huang, Ruobing Zhang, Zhiwei Xiong. *IEEE TMI, 2024.*

[12] **Learning Multiscale Consistency for Self-Supervised Electron Microscopy Instance Segmentation**
Yinda Chen, Wei Huang, Xiaoyu Liu, Shiyu Deng, Qi Chen, Zhiwei Xiong. *ICASSP, 2024.*

[13] **Self-supervised Neuron Segmentation with Multi-agent Reinforcement Learning**
Yinda Chen, Wei Huang, Shenglong Zhou, Qi Chen, Zhiwei Xiong. *IJCAI, 2023.*

Conference Abstracts and Presentations

Connectome-scale chemical lateralization predicts asymmetric memory computation in the adult Drosophila mushroom body

SfN 2026; poster abstract accepted.

Vision-Language Pretraining Using Generative Methods

Invited talk, JD AI Team, 2023.03.

Research Funding and Honors

NSFC Young Students Basic Research Program (Doctoral Student) 2025–2027

Principal Investigator; awarded in 2024 for the 2025–2027 project period; RMB 300,000.

CAST Doctoral Student Program; cultivating society: CCF 2026

Zenghua Scholarship, University of Science and Technology of China 2026

Interdisciplinary Contest in Modeling (ICM), Outstanding Winner 2024

National Graduate Scholarship 2022

Xiamen University Academic Star 2021

National College Student Mathematics Competition (Non-Mathematics), Second Prize in the Final 2021

Jingrun Cup Mathematics Competition (Mathematics Majors), First Place, Xiamen University 2021

Internet+ Innovation and Entrepreneurship Competition, Fujian Gold Award 2021

Challenge Cup Academic Competition, Fujian First Prize 2021

National College Student Mathematics Competition (Non-Mathematics), First Place in Fujian 2020

Academic Service and Technical Skills

Reviewing: CVPR 2025, ICCV 2025, WACV 2026, NeurIPS 2024, ICML 2025, ICLR 2024, MICCAI 2025, ACM MM 2024, AISTATS 2024; IJCV and IEEE TIP.

Programming and tools: Python, C/C++, MATLAB, Java, LaTeX; PyTorch, TensorFlow, DeepSpeed, DDP; Git, Docker, CUDA/HPC.

Training systems: thousand-GPU distributed training; 100-GPU H200 and 64-GPU A40 environments.

Languages: Chinese (native); English (TOEFL 110, GRE 328).